INCH-POUND

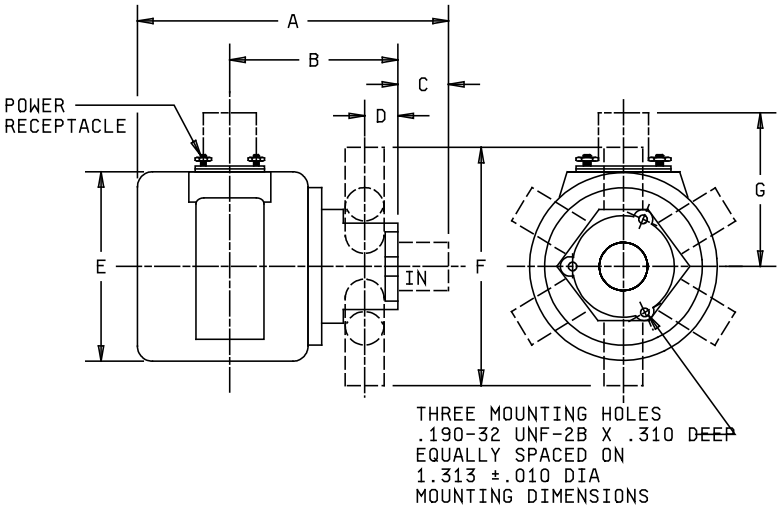
MIL-DTL-3928/7E
15 February 2011
SUPERSEDING
MIL-DTL-3928/7D
15 August 2001

DETAIL SPECIFICATION SHEET

# SWITCHES, RADIO-FREQUENCY
# TRANSMISSION LINE (COAXIAL) (MANUALLY AND SOLENOID OPERATED)
# CLASSES 1 (TYPE N CONNECTORS) AND 7 (TYPE BNC CONNECTORS)

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

Requirements for acquiring the switch described herein
shall consist of this specification sheet and MIL-DTL-3928.



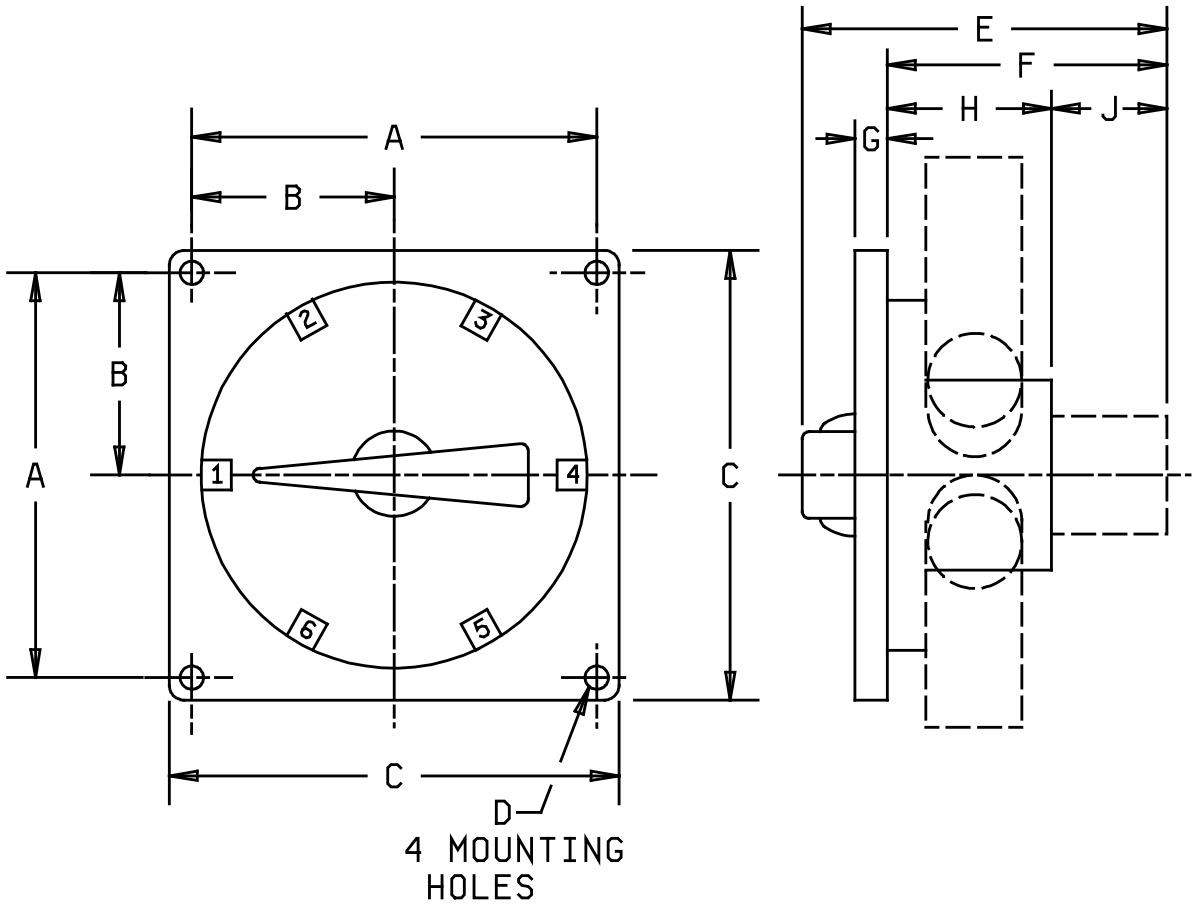

| Dimensions | | | | |
|---|---|---|---|---|
| Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min |
| A | 4.520 | 4.500 | 114.81 | 114.30 |
| B | 2.260 | 2.210 | 57.40 | 56.13 |
| C | .760 | .680 | 19.30 | 17.27 |
| D | .700 | .600 | 17.78 | 15.24 |
| E | 2.210 | 2.190 | 56.13 | 55.63 |
| F | 3.010 | 2.990 | 76.45 | 75.95 |
| G | 2.260 | 2.240 | 57.40 | 56.90 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals.

FIGURE 1. Dimensions and configuration for PIN M3928/7-11.



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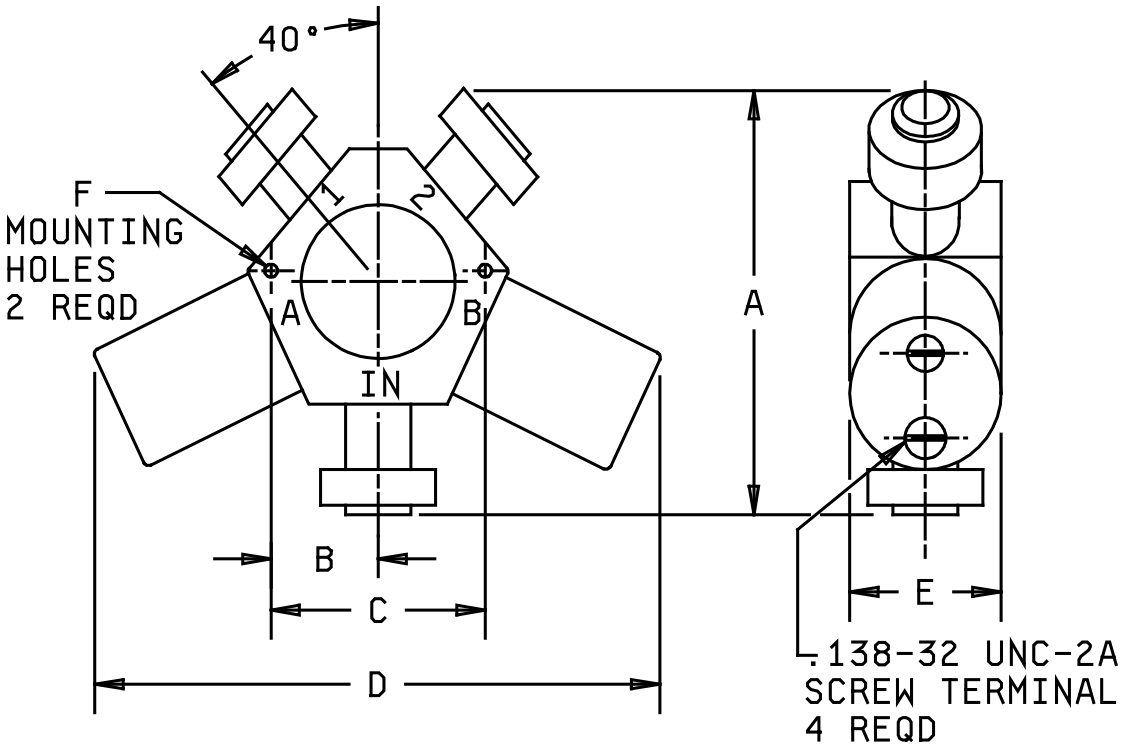

| Dimensions | | | | |
|---|---|---|---|---|
| Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min |
| A | 2.510 | 2.490 | 63.75 | 63.25 |
| B | 1.260 | 1.240 | 32.00 | 31.50 |
| C | 3.135 | 3.115 | 79.63 | 79.12 |
| D | .228 dia | .208 dia | 5.79 dia | 5.28 dia |
| E | 3.510 | 3.490 | 89.15 | 88.65 |
| F | 2.59 | 2.53 | 65.79 | 64.26 |
| G | .170 | .150 | 4.32 | 3.81 |
| H | 1.84 | 1.78 | 46.74 | 45.21 |
| J | .760 | .740 | 19.30 | 18.80 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.76 mm) for two place decimals.

FIGURE 2. Dimensions and configuration for PINs M3928/7-02, M3928/7-06, M3928/7-12, M3928/7-13, and M3928/7-22.



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POSITION 1 - NORMALLY OPEN
POSITION 2 - NORMALLY CLOSED
POSITION IN - COMMON

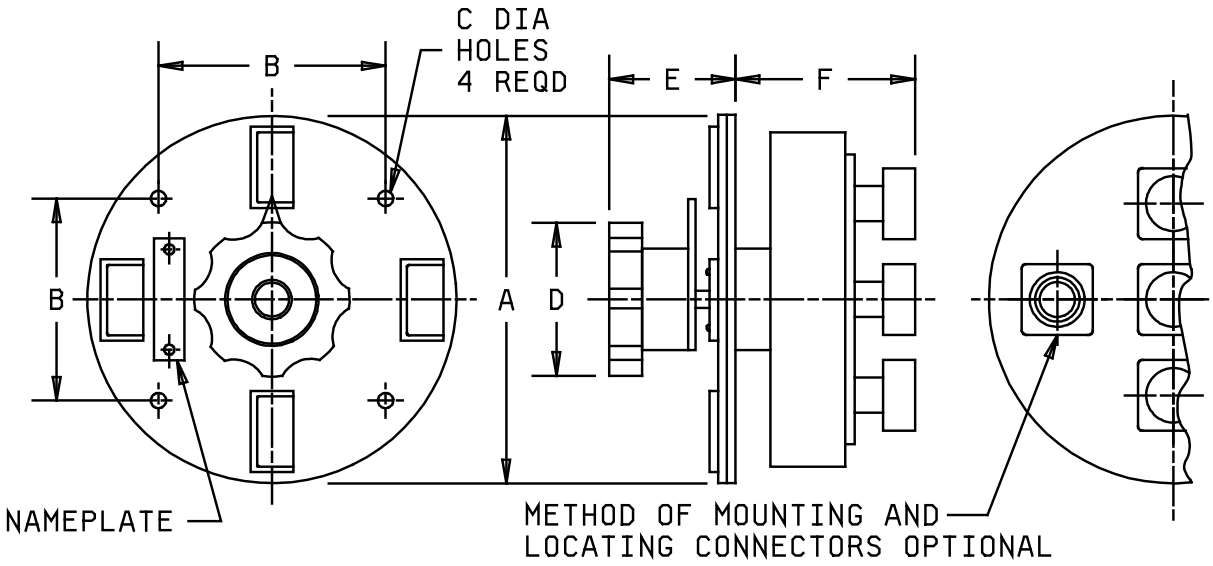
| Dimensions | | | | |
|---|---|---|---|---|
| Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min |
| A | 2.410 | 2.390 | 61.20 | 60.70 |
| B | .558 | .538 | 14.17 | 13.67 |
| C | 1.106 | 1.086 | 28.09 | 27.58 |
| D | 4.260 | 4.240 | 108.20 | 107.70 |
| E | .875 | .855 | 22.22 | 21.72 |
| F | .150 dia | .130 dia | 3.81 dia | 3.30 dia |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.76 mm) for two place decimals.

FIGURE 3. Dimensions and configuration for PINs M3928/7-09, M3928/7-10, M3928/7-17, M3928/7-18, M3928/7-19, and M3928/7-20.



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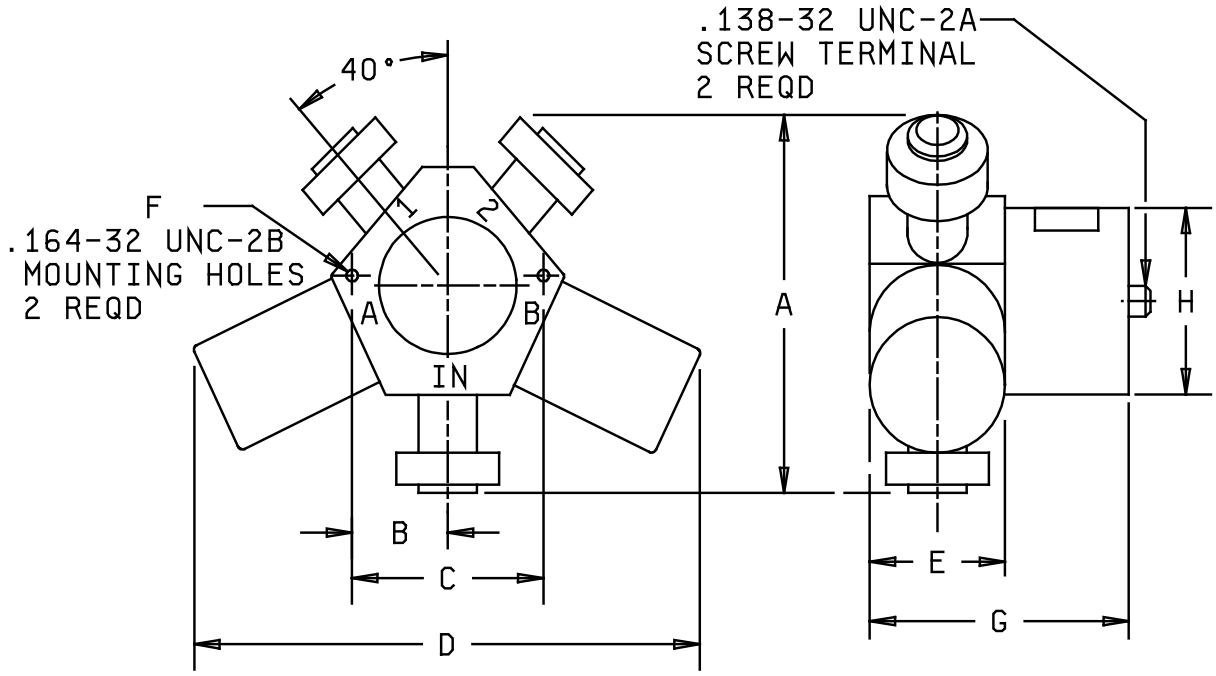

| Dimensions | | | | |
|---|---|---|---|---|
| Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min |
| A | 5.010 | 4.990 | 127.25 | 126.75 |
| B | 2.510 | 2.490 | 63.75 | 63.25 |
| C | .198 | .178 | 5.03 | 4.52 |
| D | 2.410 | 2.390 | 61.21 | 60.71 |
| E | 1.300 | 1.280 | 33.02 | 32.51 |
| F | 3.010 | 2.990 | 76.45 | 75.95 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals.

FIGURE 4. Dimensions and configuration for PINs M3928/7-14, M3928/7-15, and M3928/7-16.



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POSITION 1 - NORMALLY OPEN
POSITION 2 - NORMALLY CLOSED
POSITION IN - COMMON

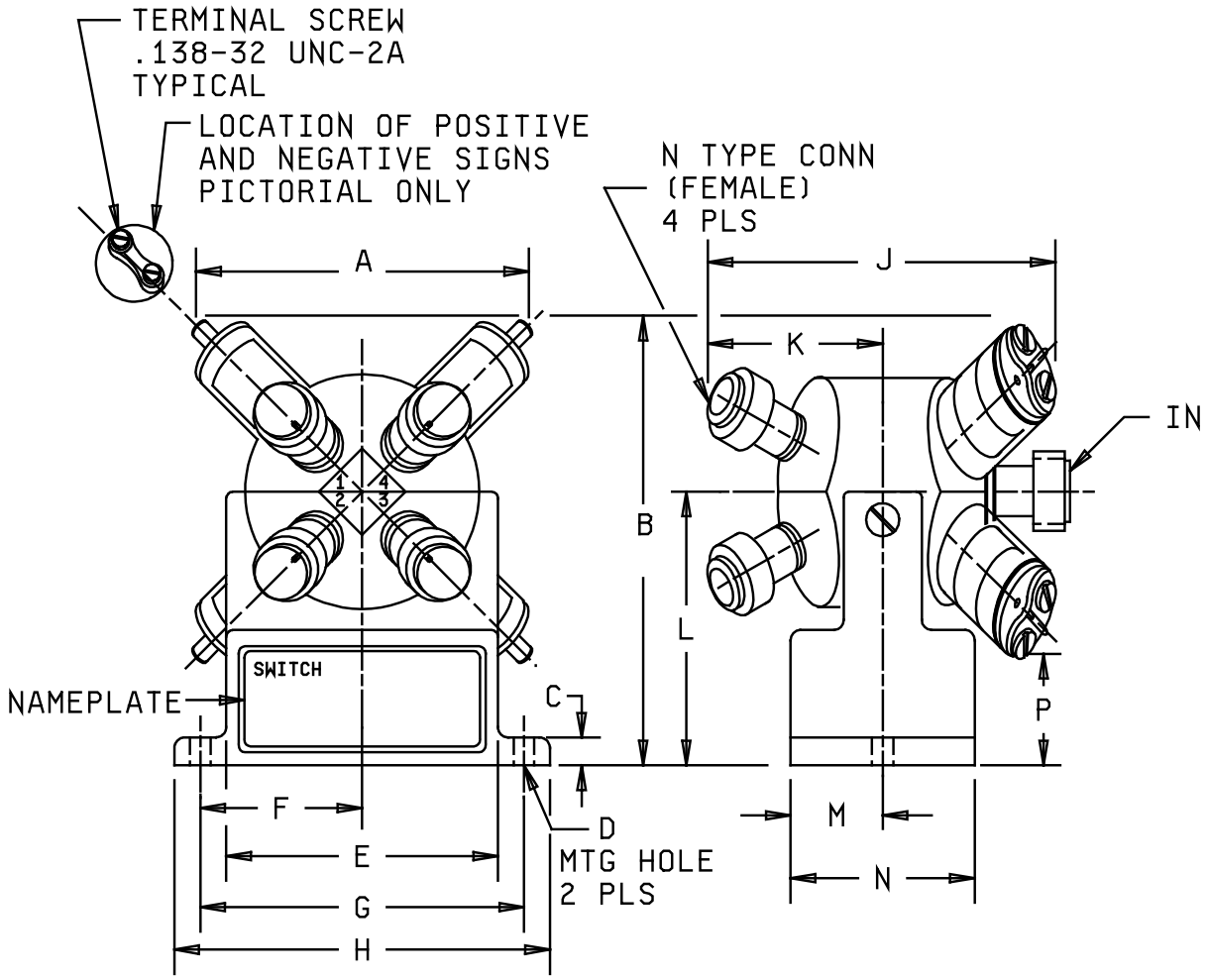
| Dimensions | | | | |
|---|---|---|---|---|
| Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min |
| A | 2.410 | 2.390 | 61.21 | 60.71 |
| B | .558 | .538 | 14.17 | 13.67 |
| C | 1.106 | 1.086 | 28.09 | 27.58 |
| D | 3.62 | 3.56 | 91.9 | 90.4 |
| E | .875 | .855 | 22.22 | 21.72 |
| F | .260 deep | .240 deep | 6.60 deep | 6.10 deep |
| G | 1.69 | 1.63 | 42.9 | 41.4 |
| H | 1.31 | 1.25 | 33.3 | 31.8 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.76 mm) for two place decimals.

FIGURE 5. Dimensions and configuration for PIN M3928/7-23.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:15Z
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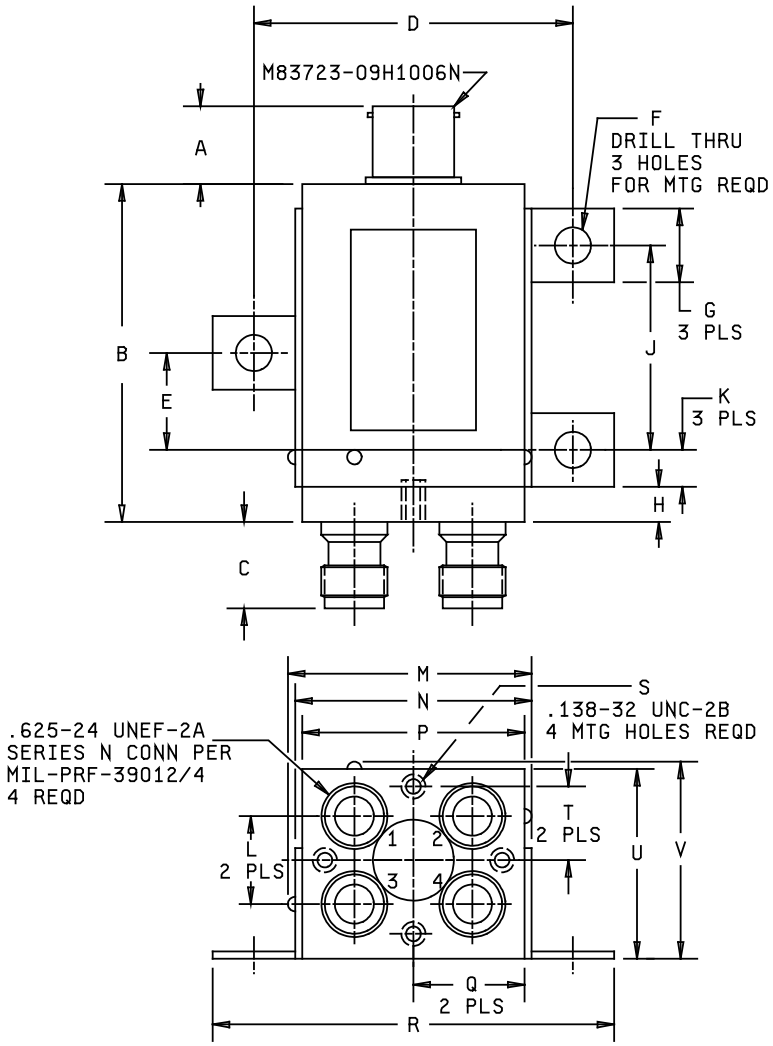

| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Letter | Inches | | Millimeters | | Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min | | Max | Min | Max | Min |
| A | 2.87 | --- | 72.9 | --- | H | 3.06 | 3.00 | 77.7 | 76.2 |
| B | 3.43 | --- | 87.1 | --- | J | 2.81 | --- | 71.4 | --- |
| C | .28 | .22 | 7.1 | 5.6 | K | 1.09 | 1.03 | 27.7 | 26.2 |
| D | .213dia | .193 dia | 5.41 dia | 4.90 dia | L | 2.02 | 1.96 | 51.3 | 49.8 |
| E | 1.87 | 1.81 | 47.5 | 46.0 | M | .56 | .50 | 14.2 | 12.7 |
| F | 1.260 | 1.240 | 32.00 | 31.50 | N | 1.06 | 1.00 | 26.9 | 25.4 |
| G | 2.510 | 2.490 | 63.75 | 63.25 | P | --- | .50 | --- | 12.7 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.76 mm) for two place decimals.

FIGURE 6. Dimensions and configuration for PIN M3928/7-24.



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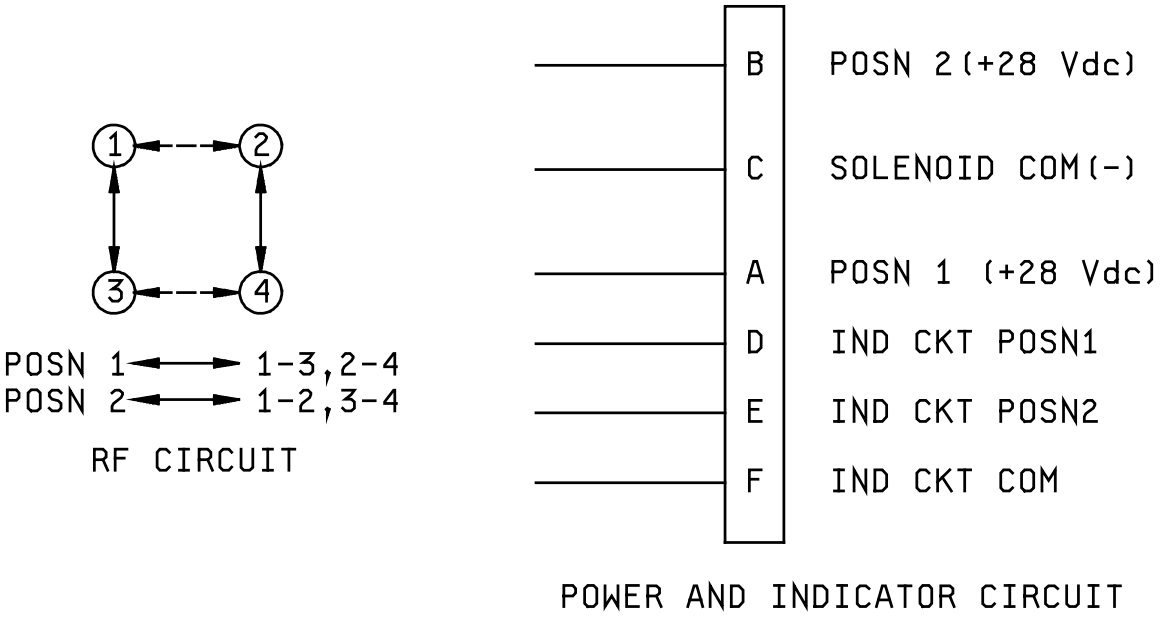

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.76 mm) for two place decimals.

FIGURE 7. Dimensions and configuration for PINs M3928/7-25 and M3928/7-26.



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| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Letter | Inches | | Millimeters | | Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min | | Max | Min | Max | Min |
| A | .49 | --- | 12.4 | --- | L | .895 | .875 | 22.73 | 22.23 |
| B | 2.60 | --- | 66.0 | --- | M | 2.04 | --- | 51.8 | --- |
| C | .65 | --- | 16.5 | --- | N | 1.95 | --- | 49.5 | --- |
| D | 2.610 | 2.590 | 66.29 | 65.79 | P | 1.78 | --- | 45.2 | --- |
| E | .760 | .740 | 19.30 | 18.80 | Q | .885 | .865 | 22.48 | 21.97 |
| F | .198dia | .192 dia | 5.03 dia | 4.88 dia | R | 3.28 | 3.22 | 83.3 | 81.8 |
| G | .63 | .57 | 16.0 | 14.5 | S | .28 deep | .22 deep | 7.1 deep | 5.6 deep |
| H | .28 | .22 | 7.1 | 5.6 | T | .730 | .710 | 18.54 | 18.03 |
| J | 1.510 | 1.490 | 38.35 | 37.85 | U | 1.78 | --- | 45.2 | --- |
| K | .33 | .27 | 8.4 | 6.9 | V | 1.87 | --- | 45.7 | --- |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.76 mm) for two place decimals.
4. Remove all burrs and sharp edges .015 inch (0.38 mm) maximum.
5. All machined surfaces shall be in accordance with ASME-B46.1.

FIGURE 7. Dimensions and configuration for PINs M3928/7-25 and M3928/7-26 - Continued.



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TABLE I. Electrical and performance characteristics.

| PIN M3928/7 | Figure No. | Housing | Manual or remote | Solenoid or motor | Fail safe or latching | RF conn type | Frequency range dc to GHz | VSWR | Insertion loss (dB) | Isolation (dB) | Config. | Position indicating circuit (and current rating) | Life cycles x 1000 | Nominal oper voltage | Oper current (A) | Pick-up voltage | Drop-out voltage | Switch time (ms) | Power connector | Weight (lbs) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | max | max | min | | | | | max | | | | | max |
| 02N, S | 2 | I | M | --- | --- | N | 10 | 1.5 | 0.52 | 25 | 1P2T | None | 25 | --- | --- | --- | --- | --- | --- | 1 |
| 06N, S | 2 | I | M | --- | --- | N | 10 | 1.5 | 0.52 | 25 | 1P6T | None | 25 | --- | --- | --- | --- | --- | --- | 1.25 |
| 09N, S | 3 | I | R | S | F | N | 10 | 1.5 | 0.52 | 20 | 1P2T | None | 1000 | 110 dc | 2 | 60 | 2 | 20 | Screw 6-32NC-2A | .5 |
| 10N, S | 3 | I | R | S | F | BNC | 10 | 1.5 | 0.52 | 25 | 1P2T | None | 1000 | 26 dc | 3 | 18 | 0.5 | 20 | Screw 6-32NC-2A | .5 |
| 11N, S | 1 | I | R | S | --- | N | 10 | 1.5 | 0.52 | 25 | 1P3T | None | 100 | 110 dc | 3 | 60 | 2 | 250 | MS3456-14S-2S | 2 |
| 12N, S | 2 | I | M | --- | --- | N | 10 | 1.5 | 0.52 | 25 | 1P3T | None | 25 | --- | --- | --- | --- | --- | --- | 1 |
| 13N, S | 2 | I | M | --- | --- | BNC | 10 | 1.5 | 0.52 | 25 | 1P6T | None | 25 | --- | --- | --- | --- | --- | --- | 1.25 |
| 14N, S | 4 | I | M | --- | --- | N | 10 | 1.5 | 0.52 | 25 | 1P2T | None | 25 | --- | --- | --- | --- | --- | --- | 1.32 |
| 15N, S | 4 | I | M | --- | --- | N | 10 | 1.5 | 0.52 | 25 | 1P6T | None | 25 | --- | --- | --- | --- | --- | --- | 1.5 |
| 16N, S | 4 | I | M | --- | --- | N | 10 | 1.5 | 0.52 | 25 | 1P4T | None | 25 | --- | --- | --- | --- | --- | --- | 1.5 |
| 17N,S | 3 | I | R | S | F | N | 10 | 1.5 | 0.52 | 20 | 1P2T | None | 1000 | 26 dc | 2 | 18 | 0.5 | 20 | Screw 6-32NC-2A | .5 |
| 18N, S | 3 1/ | I | R | S | F | N | 10 | 1.5 | 0.52 | 25 | 1P2T | None | 1000 | 26 dc | 2 | 18 | 0.5 | 20 | Screw 6-32NC-2A | .5 |
| 19N, S | 3 1/ | I | R | S | F | N | 10 | 1.5 | 0.52 | 25 | 1P2T | None | 1000 | 110 dc | 2 | 60 | 2 | 20 | Screw 6-32NC-2A | .5 |
| 20N, S | 3 2/ | I | R | S | F | N | 10 | 1.5 | 0.52 | 25 | 1P2T | None | 1000 | 28 dc | 2 | 18 | 0.5 | 20 | Screw 6-32NC-2A | .5 |
| 22N, S | 2 | I | M | --- | --- | N | 10 | 1.5 | 0.52 | 25 | 1P4T | None | 25 | --- | --- | --- | --- | --- | --- | 1 |
| 23N, S | 5 | I | R | S | F | BNC | 10 | 1.5 | 0.52 | 25 | 1P2T | None | 1000 | 115 V ac 3/ | 2 | 93 | 5 | 20 | Screw 6-32NC-2A | .5 |
| 24N, S | 6 | I | R | S | --- | N | 10 | 1.5 | 0.52 | 25 | 1P4T | None | 1000 | 28 V dc | 2 | 18 | 0.5 | 20 | Screw 6-32NC-2A | .75 |
| 25N, S | 7 | I | R | S | L | N | 12.4 | 1.5 | 0.5 | 60 | TR | None | 100 | 28 V dc | 0.64 | 20 | --- | 20 | MS83723-13R1006N | .75 |
| 26N, S | 7 | I | R | S | L | N | 12.4 | 1.5 | 0.5 | 60 | TR | Yes 4A res 2A ind | 100 | 28 V dc | 0.64 | 20 | --- | 20 | MS83723-13R1006N | .75 |

1/ Both positions normally open.
2/ Both positions normally closed.
3/ Frequency 400 Hz.



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REQUIREMENTS:

Dimensions and configurations: See figure 1 through figure 7.

Electrical and performance characteristics: See table I.

RF power handling capability (average): 100 watts (minimum).

RF connectors: Series N connectors shall mate with Series N male connectors in accordance with MIL-PRF-39012/1. BNC connectors shall mate with BNC male connectors in accordance with MIL-PRF-39012/16.

Nominal impedance: 50 ohms.

Termination: Open.

Vibration: Method I.

Operating temperature: -55°C to +85°C.

Part or Identifying Number (PIN): M3928/7- (and dash number from table I).

TABLE II. PIN to type cross-reference. 1/

| PIN M3828/7 | Type SA- /U | PIN M3828/7 | Type SA- /U |
|---|---|---|---|
| 02 | 1321 | 17 | --- |
| 06 | 1325 | 18 | --- |
| 09 | 1328 | 19 | --- |
| 10 | 1329 | 20 | --- |
| 11 | 1330 | 22 | --- |
| 12 | 1331 | 23 | --- |
| 13 | 1332 | 24 | --- |
| 14 | 273 | 25 | --- |
| 15 | 275 | 26 | --- |
| 16 | 274 | | |

1/ PINs M3928/7-01, M3928/7-03, M3928/7-04, M3928/7-05, M3928/7-07, M3928/7-08, and M3928/7-21 have been deleted.



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Referenced documents: In addition to MIL-DTL-3928, this document references the following
MIL-PRF-39012/1
MIL-PRF-39012/4
MIL-PRF-39012/16
ASME-B46.1

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2011-003)

Review activities:
Army - AV, MI
Navy - MC
Air Force - 99

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